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PERFORMANCE OF THE MANCHESTER TRIAGE SYSTEM IN ADULT MEDICAL EMERGENCY PATIENTS: A PROSPECTIVE COHORT STUDY

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Abstract—Background: Accurate initial patient triage in the emergency department (ED) is pivotal in reducing time to effective treatment by the medical team and in expediting patient flow. The Manchester Triage System (MTS) is widely implemented for this purpose. Yet the overall effectiveness of its performance remains unclear. **Objectives:** We investigated the ability of MTS to accurately assess high treatment priority and to predict adverse clinical outcomes in a large unselected population of medical ED patients. **Methods:** We prospectively followed consecutive medical patients seeking ED care for 30 days. Triage nurses implemented MTS upon arrival of patients admitted to the ED. The primary endpoint was high initial treatment priority adjudicated by two independent physicians. Secondary endpoints were 30-day all-cause mortality, admission to the intensive care unit (ICU), and length of stay. We used regression models with area under the receiver operating characteristic curve (AUC) as a measure of discrimination. **Results:** Of the 2407 patients, 524 (21.8%) included patients (60.5 years, 55.7% males) who were classified as high treatment priority; 3.9% (n = 93) were transferred to the ICU; and 5.7% (n = 136) died. The initial MTS showed fair prognostic accuracy in predicting treatment priority (AUC 0.71) and ICU admission (AUC 0.68), but not in predicting mortality (AUC 0.55). Results were robust across most predefined subgroups, including patients diagnosed with

infections, or cardiovascular or gastrointestinal diseases. In the subgroup of neurological symptoms and disorders, the MTS showed the best performance. **Conclusion:** The MTS showed fair performance in predicting high treatment priority and adverse clinical outcomes across different medical ED patient populations. Future research should focus on further refinement of the MTS so that its performance can be improved. **Trial Registration:** Clinicaltrials.gov: NCT01768494. © 2015 Elsevier Inc.

Keywords—Manchester triage system; triage; emergency department; injury severity; mortality

INTRODUCTION

Hospital emergency departments (EDs) are increasingly overwhelmed by patients with urgent and non-urgent problems (1,2). This leads to crowded waiting rooms with inordinately long waiting times for many patients. As a consequence, patients needing urgent care may not be treated in time, whereas patients with non-urgent complaints may receive expensive emergency care unnecessarily. The time a patient must wait to receive

effective treatment is one of the most important predictors of outcomes for patients affected with serious medical conditions such as septicemia, pneumonia, stroke, and myocardial infarction (3–6). For this reason, a well-validated and accurate triage tool in the ED is pivotal for optimal initial triage.

The word *triage* is derived from the French verb *trier*, which means to sort, separate, and select. Triage means the rapid, standardized initial assessment and prioritization of patients arriving at the ED. Various triage scores have been devised to improve standardization of this important process. The triage scores categorize patients into priority groups according to the severity of their condition until a patient can be examined by the medical team. Time is crucial in triage: Wait times vary according to scores of perceived urgency; for example, the most severe cases are transported for medical examination immediately. Moreover, if limited resources lead to overcrowding in the ED, triage scores aim to optimize workloads in the ED. In such circumstances, medically urgent or life-threatening conditions should be prioritized (7,8). Accordingly, patients with life-threatening injuries or illnesses must be accurately identified within minutes of arrival (8,9).

Several five-level triage-scoring systems are currently used. They include the Australasian Triage Scale, the Canadian Triage and Acuity Scale, the Manchester Triage System (MTS), and the Emergency Severity Index (8,10,11). The MTS was developed in the United Kingdom by the Manchester Triage Group in 1994/1995 (12). In 2008 the first German translation of the second edition was released, and since then it has found widespread application in EDs in German-speaking areas. Among these scoring systems, the MTS is currently the most widely used in European and North American health care settings (8). Yet only a few studies have focused on validating the clinical performance of the MTS in unselected medical ED patients. In contrast, our aim was to study the real-time performance of the MTS in a large unselected population of medical emergency patients with regard to triage priority and adverse clinical outcomes.

MATERIALS AND METHODS

Study Design and Setting

This is a prospective, observational cohort study. We included consecutive medical patients seeking care in a tertiary care hospital ED in Switzerland between April 2013 and October 2013, except for non-adult and nonmedical patients.

The study was approved by the Institutional Review Board of the Canton of Aargau (approval number EK

2012/059) and registered at the ClinicalTrials.gov registration Web site (NCT01768494). The requirement for informed consent was waived due to the observational and quality control character of the study.

We recorded initial vital signs of all patients when they presented at the ED. Clinical information including socio-demographics, comorbidities, and patient outcomes were assessed prospectively until hospital discharge using the routinely gathered information from the hospital electronic medical system used for coding of Diagnosis-Related Groups. This available information supported the reliable assessment of baseline characteristics such as demographics, comorbidities, acute medical conditions requiring the ED visit, and different patient outcomes, including in-hospital mortality, resource use (e.g., admission to the intensive care unit [ICU] and length of stay [LOS] in the hospital). The data were verified and completed by hand on the basis of discharge reports.

Definition of Patient Classification

Patients were grouped according to their presenting chief complaint (thoracic pain, other pain, neurological symptoms, respiratory symptoms, blood loss, diarrhea, vomitus, dysuria, fever, other) and on the basis of the main ED diagnosis (infection, cardiovascular, metabolic, cancer, neurological, gastrointestinal, pulmonary, other).

MTS Assessment

Our study is based on the German version of MTS, which differs in several respects from the English version. Initial MTS was calculated upon ED admission by trained triage nurses. In brief, the MTS consists of 50 predefined flowcharts, each presenting a complaint such as “abdominal pain in adults,” “allergy,” or “diarrhea and vomiting.” As recommended, triage nurses used presentation diagrams, which best described the patient’s condition. On the basis of these diagrams, the patient was assigned to one of five priority categories, which indicate the maximum time until a patient is seen by a physician—Red: immediate care, no wait time; Orange: very urgent, 10-min wait time; Yellow: urgent, 30-min wait time; Green: normal, standard, 90-min wait time; Blue: non-urgent, 120-min wait time (13).

Endpoints and Outcome Assessment

The primary endpoint of this study was to determine the adjudication of treatment priority. Similar to other validation studies, in our study, two independent physicians with training in internal medicine and emergency medicine retrospectively evaluated what the real degree of urgency (gold standard) would have been, based on all

clinical data, results of diagnostic tests, and the final diagnosis that was available at ED discharge (8,14–16). Specifically, the main question for the adjudicators was “*under difficult circumstances, what is the maximum possible time that this patient would have had to wait before being seen by a physician?*” Adjudicators used the same five MTS time categories and evaluation was blinded with regard to the initial MTS, as well as to the conclusions of the other physicians. We used this gold standard as a continuous variable and also dichotomized this outcome variable by collapsing the initial five priority categories into two categories: high vs. low priority. Codes “Red” (patient could not wait) and “Orange” (10 min) were grouped together into a “high priority” cluster; whereas codes “Yellow” (30 min), “Green” (90 min), and “Blue” (2 h) were grouped together into a “low priority” cluster. To standardize the adjudication process, we provided physicians with a reference list of illnesses and associated MTS classifications as published elsewhere (17). If the two adjudicators disagreed, a third independent physician reviewed the case until consensus was reached among the three adjudicators.

Secondary endpoints included 30-day all-cause mortality, ICU admission, and LOS. We used data from electronic medical records, and all patients were contacted by telephone 30 days after hospital admission to assess vital and functional status.

Statistical Analysis

Our aim was to validate the performance of the MTS for identifying patients at high treatment priority defined as adjudicated treatment priority (primary endpoint) and 30-day all-cause mortality, ICU admission, and LOS (secondary endpoints). Categorical variables were expressed as counts (percentage), and continuous variables as medians and interquartile ranges (IQR). If applicable, 95% confidence interval (CI) was provided.

We used regression models with area under the receiver operating characteristic (ROC) curve to assess associations of baseline parameters and outcomes. These are determined by plotting the true-positive rate (sensitivity) against the false-positive rate (1–specificity) associated with a variable’s values. Area under the curve (AUC) indicates the probability of that variable correctly categorizing an individual, here as a high priority patient. Higher AUC reflects greater accuracy: 0.5, the null value, indicates “coin-toss accuracy”; 1.0, the maximum value, indicates infallibility.

We also report odds ratio (OR) from logistic regression analysis as a measure of the association between an exposure and an outcome. It represents the odds that an outcome will occur (e.g., ICU admission) given a

particular exposure (e.g., high priority), compared to the odds of the outcome occurring in the absence of that exposure (e.g., low priority). In a second step, subgroup analysis was performed to investigate the performance of the MTS within predefined patient populations based on different presenting chief complaints and main diagnoses as described above.

We used STATA 12.1 (Stata Corp, College Station, TX). All testing was two-tailed, with $p < 0.05$ considered as indicating statistical significance.

RESULTS

Patient Population

Overall, 2407 medical patients were included; 55.7% were male, with a median age of 60.5 years (range 16–97 years). Baseline characteristics of the overall cohort and stratified according to MTS category are presented in Table 1. The two leading reasons for seeking medical assistance were pain (except for thoracic pain [40.4% had thoracic and non-thoracic pain] and neurological symptoms [22.6%]). The main diagnoses at ED admission were neurological (26.4%) and cardiovascular (25.2%). Patients had a high burden of comorbidities including hypertension (42%), cancer (16%), renal failure (15.4%), diabetes (14.9%), and coronary artery disease (12.3%).

Association of MTS with Adjudicated Triage Priority

A total of 524 (21.8%) patients were classified as high treatment priority patients (categories Red and Orange). Figure 1 shows the number of patients with correct and incorrect MTS classification compared to the adjudicated gold-standard definition. Overall, there was a moderate concordance, except for a high percentage of disagreement in the Orange category (58.9%). Showing a fair discrimination (AUC 0.71), the MTS association with high treatment priority was significant (OR 2.8, 95% CI 2.43–3.20; $p < 0.001$) (Table 2). Analyses were performed to predict urgency, sensitivity, and specificity (Table 3).

As shown in Figure 2, we evaluated the MTS in different predefined subgroups. We found heterogeneity with nonsignificant results in the subgroup of patients with chief complaints of fever, diarrhea, vomitus, and dysuria, but best results in patients with neurological symptoms, blood loss, and cardiovascular diseases.

Association of MTS and 30-Day Prediction of Mortality

All-cause mortality within 30 days was 5.7% ($n = 136$). There was a stepwise increase in mortality with higher

Table 1. Baseline Characteristics of Patients

Characteristics	MTS Categories					Overall	p-Value
	Red	Orange	Yellow	Green	Blue		
Number of patients	21 (0.87%)	621 (25.80%)	976 (40.55%)	722 (30.0%)	67 (2.78%)	2407	
Demographic characteristics							
Age, median (IQR)	71 (56–80)	67 (54–77)	63 (47–74)	62 (46–75)	58 (36–75)	60.50 (48–75.3)	<0.001
Male, n (%)	13 (61.9%)	378 (60.9%)	551 (56.5%)	370 (51.2%)	28 (41.8%)	1340 (55.7%)	<0.001
Outcomes							
Physician's assessment urgent, n (%)	17 (81.0%)	255 (41.1%)	188 (19.3%)	58 (8.0%)	6 (9.0%)	524 (21.8%)	<0.001
In-hospital death, n (%)	4 (19.0%)	25 (4.0%)	30 (3.1%)	14 (1.9%)	2 (3.0%)	75 (3.1%)	<0.001
30-day mortality, n (%)	4 (19.0%)	42 (6.8%)	52 (5.3%)	36 (5.0%)	2 (3.0%)	136 (5.7%)	0.037
ICU admission, n (%)	7 (33.3%)	47 (7.6%)	33 (3.4%)	6 (0.8%)	0 (0.0%)	93 (3.9%)	<0.001
LOS, mean (SD), days	5.8 (6.7)	4.6 (4.7)	4.0 (5.0)	4.0 (5.3)	3.6 (4.2)	4.2 (5.0)	0.086
Rehospitalization, n (%)	0 (0.0%)	48 (7.7%)	56 (5.7%)	47 (6.5%)	6 (9.0%)	157 (6.5%)	0.33
Clinical findings							
SO ₂ , median (IQR), (%)	95 (91–96)	96 (93–98)	96 (93.6, 97)	96 (94–97)	96 (95–98)	96 (94–97)	0.050
BPS, median (IQR), mm Hg	146 (115–156)	142 (125–162)	138 (122–153)	135 (121–153)	142 (121–154)	138 (123–155)	<0.001
BPD, median (IQR), mm Hg	79 (73–83)	83 (73–96)	81 (71–90)	80 (71–90)	83 (75–89)	81 (71–91)	<0.001
PR, median (IQR), beats/min	95 (76–122)	84 (70–101)	83 (72–96)	80 (69–90)	81.5 (65–92)	82 (70.5–95)	<0.001
T, median (IQR), °C	37 (36.2–37.4)	36.8 (36.5–37.2)	37 (36.6–37.5)	37 (36.6–37.3)	36.8 (36.5–37.2)	36.95 (36.6–37.4)	<0.001
Confusion, n (%)	12 (57.1%)	102 (16.4%)	72 (7.4%)	36 (5.0%)	3 (4.5%)	225 (9.3%)	<0.001
Discharge location, n (%)							
Home	10 (47.6%)	453 (72.9%)	829 (84.9%)	628 (87.0%)	56 (83.6%)	1976 (82.1%)	<0.001
Institution	5 (23.8%)	91 (14.7%)	83 (8.5%)	56 (7.8%)	8 (11.9%)	243 (10.1%)	
Rehabilitation	2 (9.5%)	49 (7.9%)	34 (3.5%)	24 (3.3%)	1 (1.5%)	110 (4.6%)	
Other	4 (19.0%)	28 (4.5%)	30 (3.1%)	14 (1.9%)	2 (3.0%)	78 (3.2%)	
Chief complaint, n (%)							
Pain other	4 (19.0%)	89 (14.3%)	274 (28.1%)	215 (29.8%)	20 (29.9%)	602 (25.0%)	<0.001
Thoracic pain	1 (4.8%)	186 (30.0%)	126 (12.9%)	54 (7.5%)	3 (4.5%)	370 (15.4%)	
Neurological symptoms	12 (57.1%)	189 (30.4%)	190 (19.5%)	141 (19.5%)	11 (16.4%)	543 (22.6%)	
Respiratory symptoms	2 (9.5%)	67 (10.8%)	114 (11.7%)	75 (10.4%)	8 (11.9%)	266 (11.1%)	
Blood loss	0 (0.0%)	21 (3.4%)	26 (2.7%)	12 (1.7%)	1 (1.5%)	60 (2.5%)	
Diarrhea, vomitus, dysuria	0 (0.0%)	8 (1.3%)	62 (6.4%)	73 (10.1%)	5 (7.5%)	148 (6.1%)	
Fever	0 (0.0%)	20 (3.2%)	87 (8.9%)	46 (6.4%)	2 (3.0%)	155 (6.4%)	
Other	2 (9.5%)	41 (6.6%)	97 (9.9%)	106 (14.7%)	17 (25.4%)	263 (10.9%)	
Main diagnosis, n (%)							
Infection	2 (9.5%)	52 (8.4%)	156 (16.0%)	111 (15.4%)	11 (16.4%)	332 (13.8%)	<0.001
Cardiovascular	5 (23.8%)	237 (38.2%)	233 (23.9%)	126 (17.5%)	6 (9.0%)	607 (25.2%)	
Metabolic	2 (9.5%)	5 (0.8%)	13 (1.3%)	12 (1.7%)	1 (1.5%)	33 (1.4%)	
Cancer	0 (0.0%)	6 (1.0%)	61 (6.3%)	66 (9.1%)	5 (7.5%)	138 (5.7%)	
Neurological, n (%)	10 (47.6%)	196 (31.6%)	239 (24.5%)	170 (23.5%)	20 (29.9%)	635 (26.4%)	
Gastrointestinal, n (%)	0 (0.0%)	27 (4.3%)	99 (10.1%)	101 (14.0%)	6 (9.0%)	233 (9.7%)	
Pulmonary, n (%)	0 (0.0%)	15 (2.4%)	29 (3.0%)	10 (1.4%)	0 (0.0%)	54 (2.2%)	
Other, n (%)	2 (9.5%)	83 (13.4%)	146 (15.0%)	126 (17.5%)	18 (26.9%)	375 (15.6%)	
Coexisting illnesses, n (%)							
Hypertension	11 (52.4%)	312 (50.2%)	398 (40.8%)	262 (36.3%)	28 (41.8%)	1011 (42.0%)	<0.001
CHF	3 (14.3%)	59 (9.5%)	52 (5.3%)	38 (5.3%)	3 (4.5%)	155 (6.4%)	0.003
CAD	3 (14.3%)	139 (22.4%)	98 (10.0%)	49 (6.8%)	6 (9.0%)	295 (12.3%)	<0.001

(Continued)

Table 1. Continued

Characteristics	MTS Categories					p-Value
	Red	Orange	Yellow	Green	Blue	
COPD	2 (9.5%)	42 (6.8%)	49 (5.0%)	25 (3.5%)	0 (0.0%)	0.015
Dementia	1 (4.8%)	22 (3.5%)	26 (2.7%)	17 (2.4%)	3 (4.5%)	0.60
Diabetes	7 (33.3%)	94 (15.1%)	147 (15.1%)	105 (14.5%)	5 (7.5%)	0.070
Stroke	6 (28.6%)	121 (19.5%)	69 (7.1%)	34 (4.7%)	6 (9.0%)	<0.001
Toxic	4 (19.0%)	48 (7.7%)	49 (5.0%)	35 (4.8%)	3 (4.5%)	0.009
Cancer	1 (4.8%)	65 (10.5%)	171 (17.5%)	138 (19.1%)	10 (14.9%)	<0.001
Renal failure	4 (19.0%)	83 (13.4%)	148 (15.2%)	131 (18.1%)	4 (6.0%)	0.027

MTS = Manchester Triage System; IQR = interquartile range; ICU = intensive care unit; LOS = length of stay; BPS = blood pressure systolic; BPD = blood pressure diastolic; PR = pulse rate; T = temperature; CHF = congestive heart failure; CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease.

MTS categories ($p = 0.037$) (Table 1). The MTS could significantly predict 30-day mortality (OR 1.27, 95% CI 1.03–1.56; $p = 0.027$), but with a low discrimination (AUC 0.55) (Table 2). Sensitivity as well as specificity analyses of the MTS to predict 30-day all-cause mortality are shown in Table 4. As a sensitivity analysis, we also investigated the association of MTS and 72-h mortality. Again, discrimination of MTS was low with an AUC of 0.53 for this endpoint.

As shown in Table 2 and Figure 3, subgroup analyses showed best results in the prediction of 30-day mortality for patients presenting with neurological symptoms, neurological diseases, and infections.

Association of MTS with Other Secondary Endpoints

A total of 93 (3.9%) patients were admitted to the ICU. Again, a stepwise increase in risk for ICU admission was found with higher MTS categories ($p < 0.001$) (Table 1). Overall, the MTS was significantly associated with ICU admission (OR of 3.11, 95% CI 2.31–4.19; $p < 0.001$, AUC: 0.72) (Table 2). Sensitivity as well as specificity analyses of the MTS to predict 30-day all-cause mortality are shown in Table 5.

Subgroup analysis showed strongest associations using the MTS and ICU admission prediction in patients with neurological symptoms and neurological diseases (Figure 4). Overall, mean LOS was 4.2 days. There was no significant difference in LOS across the different MTS categories ($p = 0.086$).

DISCUSSION

This prospective cohort study of adult medical patients showed, overall, a fair prognostic accuracy of the MTS at ED admission to predict treatment priority and ICU admission. There was, however, no convincing association between MTS and 30-day mortality. The absence of association between MTS and mortality arises from the immediate goal of MTS as a scoring system: The immediate goal is not to save lives but to expedite access to treatment depending on assessment of the severity of injury. However, expeditious treatment may or may not lessen mortality. Further, we looked at different predefined subgroups based on chief complaint and main initial diagnosis, in which results were robust: The best performance was manifested in patients presenting with neurological symptoms and neurological diseases. This observation suggests the possibility that MTS, and perhaps other triage scoring systems, overemphasize the severity of injury, but do not take into sufficient account the nature of presenting symptoms and complaints.

Although the MTS is being used in many EDs, there is a lack of well-collected clinical data supporting its

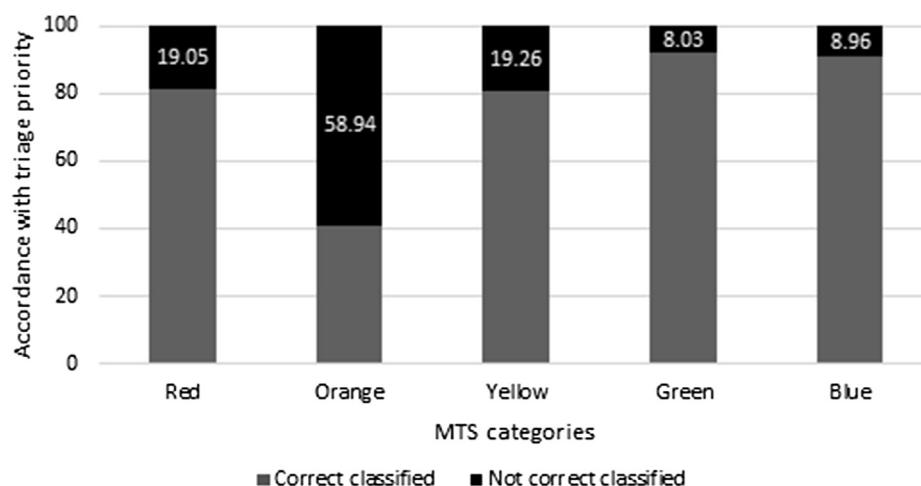


Figure 1. Manchester Triage System categories vs adjudicated triage priority. MTS = Manchester Triage System.

accuracy to predict urgency and treatment failure in unselected patients. To close this gap for the medical ED population, we were able to include a fairly large number of consecutive patients and follow them for 30 days to assess vital status and clinical outcomes. Overall, in our medical ED population, the MTS could significantly identify patients with a high treatment priority. This was particularly true in 20% of patients classified as urgent (Red) and in low urgent categories; most disagreement was found in the Orange category (58.9%). A Swedish study, however, found high interrater reliability and high accuracy of the MTS in the Red and Orange categories, whereas in the less urgent categories accuracy was lower; these findings

suggest a possible over-use of resources in low-acuity patients (18). The different findings in these two studies may have arisen because different patient populations had been studied, different methodologies had been used, or limitations might have been inherent in what was regarded as the “gold standard” for accurate assessment of treatment priority.

In a Dutch simulation study, ED nurses triaged 50 patient vignettes and triage ratings, which were later compared with the rating of MTS experts. Sensitivity for the categories Red and Orange was found to be moderate (53.2%), and specificity for the categories Yellow, Green, and Blue was high (95.1%). Nevertheless, almost

Table 2. Synoptic View of Endpoints

	Triage Priority		30-day Mortality		ICU Admission	
	OR (95% CI)	AUC	OR (95% CI)	AUC	OR (95% CI)	AUC
Overall	2.8 (2.4–3.2), $p \leq 0.001$	0.707	1.3 (1.0–1.6), $p = 0.027$	0.548	3.1 (2.3–4.2), $p \leq 0.001$	0.719
Chief complaint, n (%)						
Pain other	2.1 (1.5–3.0), $p \leq 0.001$	0.648	1.0 (0.7–1.5), $p = 0.995$	0.491	2.7 (1.5–5.0), $p = 0.001$	0.684
Thoracic pain	3.0 (1.9–4.7), $p \leq 0.001$	0.674	1.1 (0.3–3.7), $p = 0.871$	0.529	3.3 (1.4–7.8), $p \leq 0.001$	0.669
Neurological symptoms	3.9 (2.9–5.1), $p \leq 0.001$	0.763	2.7 (1.6–4.5), $p \leq 0.001$	0.715	7.5 (3.2–17.5), $p \leq 0.001$	0.817
Respiratory symptoms	2.2 (1.5–3.3), $p \leq 0.001$	0.665	1.3 (0.8–2.0), $p = 0.313$	0.543	1.6 (0.9–3.0), $p = 0.109$	0.607
Blood loss	3.5 (1.3–9.1), $p = 0.005$	0.710	NA	NA	NA	NA
Diarrhea, vomitus, dysuria	1.8 (0.7–4.5), $p = 0.221$	0.600	1.9 (0.8–4.6), $p = 0.159$	0.613	NA	NA
Fever	1.2 (0.7–2.0), $p = 0.474$	0.539	1.3 (0.5–3.2), $p = 0.548$	0.544	1.6 (0.5–5.1), $p = 0.428$	0.582
Other	3.3 (2.0–5.2), $p \leq 0.001$	0.750	0.8 (0.3–2.2), $p = 0.683$	0.458	6.2 (1.7–23.0), $p = 0.002$	0.832
Main diagnosis, n (%)						
Infection	2.7 (1.9–4.0), $p \leq 0.001$	0.690	3.1 (1.6–5.9), $p \leq 0.001$	0.715	2.2 (1.1–4.5), $p = 0.027$	0.670
Cardiovascular	3.5 (2.5–4.9), $p \leq 0.001$	0.718	1.2 (0.7–2.0), $p = 0.479$	0.531	2.9 (1.7–4.9), $p \leq 0.001$	0.684
Metabolic	2.7 (1.1–6.8), $p = 0.020$	0.722	1.2 (0.3–4.2), $p = 0.769$	0.600	1.2 (0.1–9.5), $p = 0.870$	NA
Cancer	2.2 (1.1–4.3), $p = 0.024$	0.624	1.3 (0.7–2.2), $p = 0.422$	0.531	3.6 (0.6–22.2), $p = 0.167$	0.741
Neurological	3.0 (2.4–3.8), $p \leq 0.001$	0.728	3.7 (2.0–6.6), $p \leq 0.001$	0.753	7.8 (3.3–18.6), $p \leq 0.001$	0.828
Gastrointestinal	1.7 (0.7–4.0), $p = 0.231$	0.618	1.1 (0.5–2.7), $p = 0.757$	0.528	2.0 (0.4–9.5), $p = 0.377$	0.674
Pulmonary	2.7 (0.9–7.9), $p = 0.053$	0.664	NA	0.077	1.7 (0.5–5.6), $p = 0.412$	0.588
Other	2.8 (1.9–4.1), $p \leq 0.001$	0.705	1.3 (0.5–3.4), $p = 0.563$	0.561	3.0 (1.1–7.8), $p = 0.016$	0.673

ICU = intensive care unit; OR = odds ratio; CI = confidence interval; AUC = area under the curve; NA = not applicable (due to a small number of events).

Table 3. Detailed Report of Sensitivity, and Specificity for the MTS Categories to Predict Urgency

Cutpoint	Sensitivity	Specificity	Correctly Classified	LR+	LR–
Red	3.24%	99.79%	78.77%	15.27	0.97
Orange	51.91%	80.35%	74.16%	2.64	0.60
Yellow	87.79%	38.50%	49.23%	1.43	0.32
Green	98.85%	3.24%	24.05%	1.02	0.35
Blue	100%	0%	21.77%	1	0

MTS = Manchester Triage System; LR+ = positive likelihood ratio; LR– = negative likelihood ratio.

one-third (32.9%) of the patient vignettes were rated differently by the experts: 7.6% showed overtriage and 25.3% undertriage. Undertriage in this study occurred mainly in the MTS categories Orange and Yellow—similar to our study (19). However, other studies estimated the rate of undertriage using the MTS to be significantly lower, with 5%, 11%, and 13%, respectively, in different studies (18,20,21). A systematic review of the performance of the MTS included 12 clinical studies, and concluded that MTS shows a wide interrater agreement

range; specifically, a prevalence of good and very good agreement with a low safety of MTS due to a high rate of undertriage and also low sensitivity in predicting higher urgency levels (11). In line with this meta-analysis, our results showed low sensitivity for the categories Orange and Yellow (or Red, etc.), which might suggest that urgent patient cases could be missed. Taken together with our results, the high rate of overtriage could lead to unnecessarily high use of resources in the ED, whereas undertriage may cause harm to patients.

With regard to 30-day mortality, we found low discrimination, as evidenced by an AUC of 0.55. A study by Gräff et al. came to a similar conclusion with an AUC for prediction of 30-day mortality of 0.613 (13). The following studies investigated the relationship between MTS and short-term mortality: Martins et al. showed, in a retrospective population of 316,622 patients, a clear association ($p < 0.001$) between MTS and the number of deaths (22). They demonstrated an increase in short-term mortality (defined as death occurring within the ED, usually during the first 24–48 h); moreover, the higher the level of MTS priority, the higher the number of deaths.

30-day mortality

Odds Ratio (95% CI)

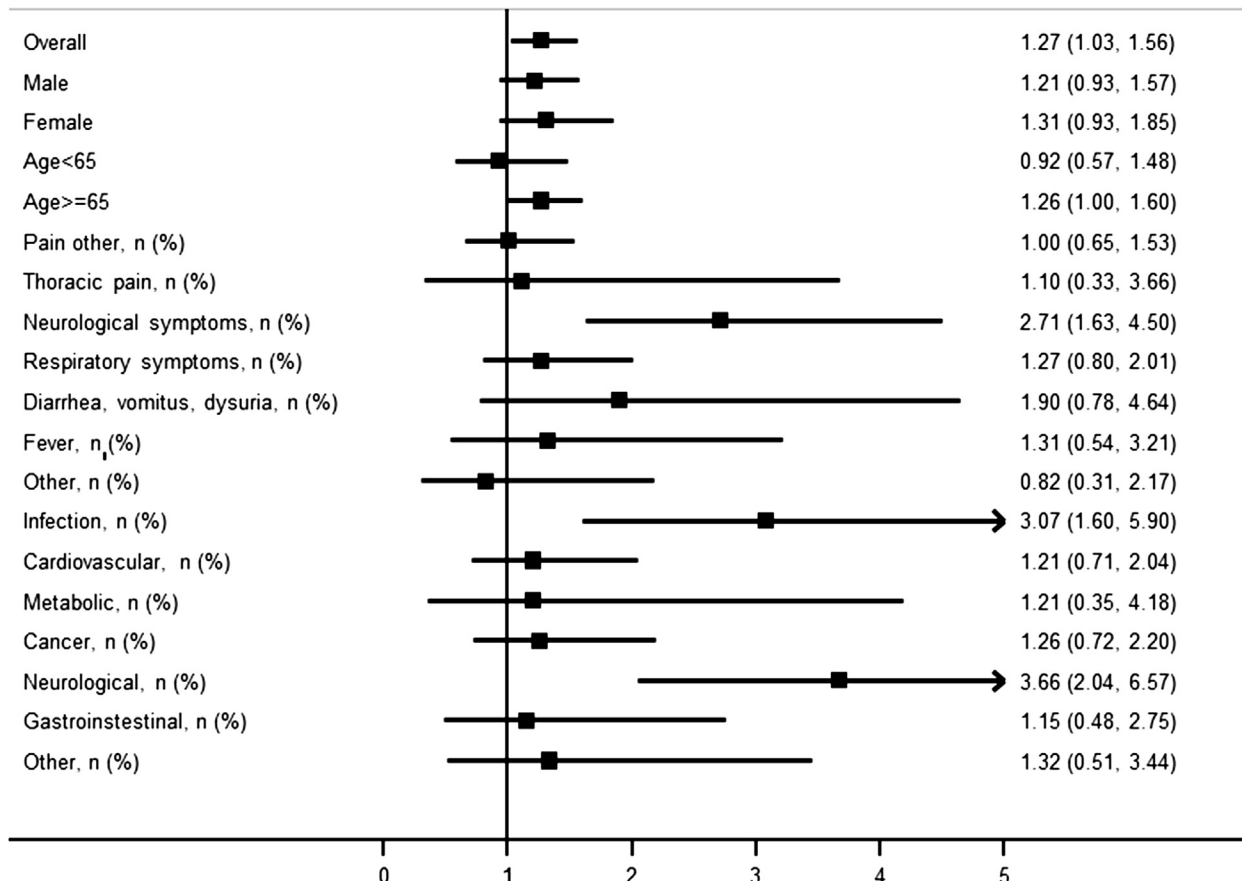


Figure 2. Forrest plot adjudicated triage priority. CI = confidence interval.

Table 4. Detailed Report of Sensitivity, and Specificity for the MTS Categories to Predict 30-Day All-cause Mortality

Cutpoint	Sensitivity	Specificity	Correctly Classified	LR+	LR–
Red	2.90%	99.30%	93.80%	3.93	0.98
Orange	33.80%	73.80%	71.50%	1.29	0.90
Yellow	72.10%	33.10%	35.30%	1.08	0.84
Green	98.50%	2.90%	8.30%	1.01	0.51
Blue	100%	0%	5.70%	1	0

MTS = Manchester Triage System; LR+ = positive likelihood ratio; LR– = negative likelihood ratio.

The death rate of patients triaged as Red was 10% (English first version) (22). Van der Wulp et al. achieved the same result but had a remarkable number of 75.9% of patients categorized as Red who died (23). Santos et al. analyzed the in-hospital death rate in a prospective study of 25,218 medical and surgical cases (24). They grouped the MTS codes into “high priority” (Red, Orange) and “low priority” (Yellow, Green, Blue) clusters.

They found the risk of death in the “high priority” cluster to be 5.58 times greater than that in the “low priority” cluster. Also, there was a clear association between priority cluster and death rate: Higher rates of death occurred in the “high priority” cluster (English second version) (24).

ICU admission rates were also found to be significantly associated with MTS priority. This finding is supported by Gräff et al., who found that 5.1% of patients who needed intensive medical treatment and ICU admission decreased from urgent to non-urgent categories (13). They described the prognostic accuracy with an ROC AUC of 0.87, which was superior to our findings of 0.72. The lower admission rate might have arisen from differences in the study populations: Gräff et al. performed their study at an interdisciplinary ED, in which all patients (except gynecologic, obstetric, and pediatric) were enrolled. The median age was 44 years. In contrast, our study included only adult medical patients, and the median age of patients enrolled was much older—63.6 years.

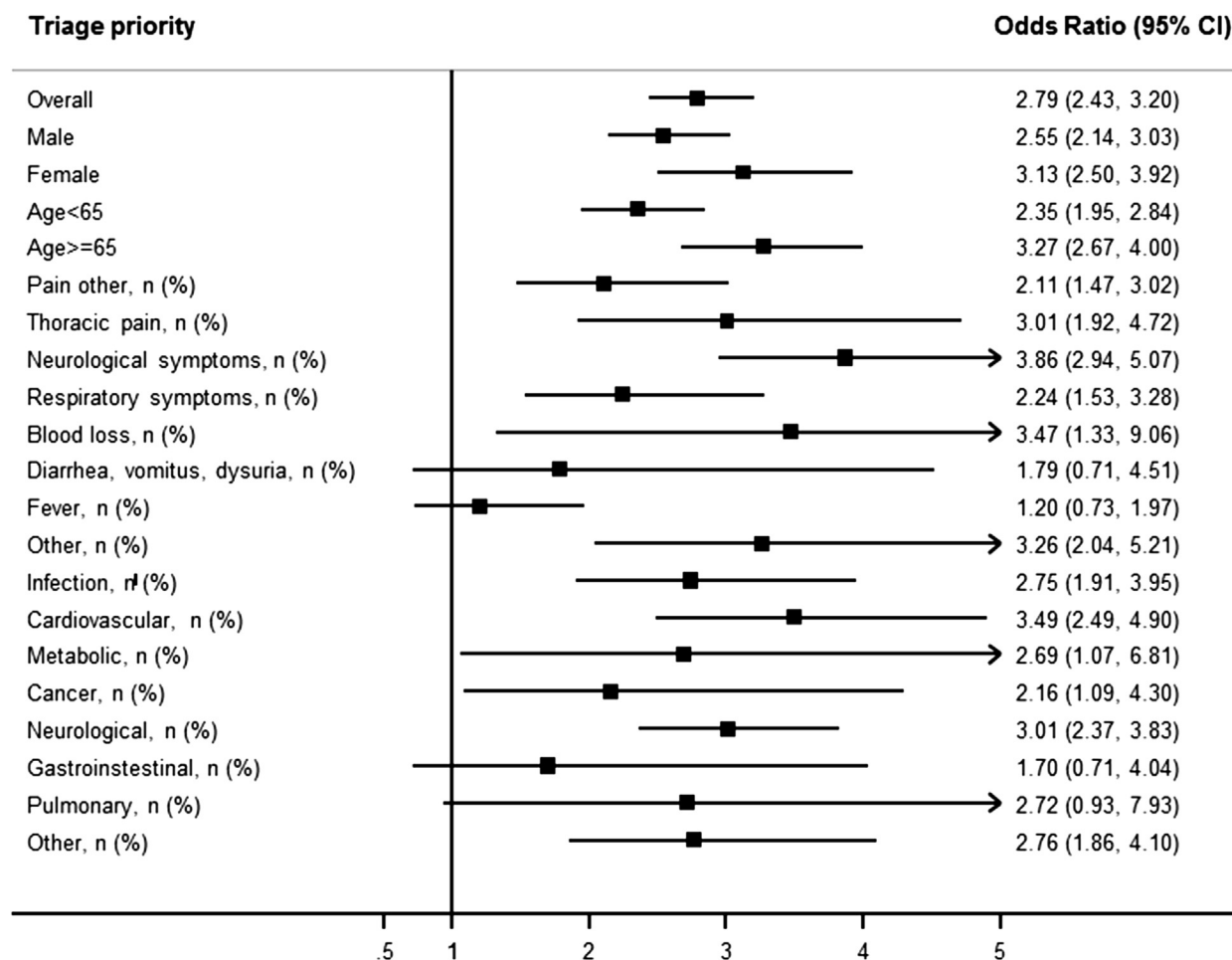


Figure 3. Forrest plot 30-day all-cause mortality. CI = confidence interval.

Table 5. Detailed Report of Sensitivity, and Specificity for the MTS Categories to Predict ICU Admission

Cutpoint	Sensitivity	Specificity	Correctly Classified	LR+	LR–
Red	7.53%	99.39%	95.85%	12.44	0.93
Orange	58.06%	74.59%	73.95%	2.29	0.56
Yellow	93.55%	33.84%	36.14%	1.41	0.19
Green	100%	2.90%	6.65%	1.03	0.00
Blue	100%	0%	3.86%	1	0

MTS = Manchester Triage System; ICU = intensive care unit; LR+ = positive likelihood ratio; LR– = negative likelihood ratio.

Finally, we found an association between MTS category and hospital LOS: Red showed the longest LOS. This association is supported by data presented by Storm-Versloot et al. as well as Gräff et al., in which statistical significance was achieved in a population of sufficient size (45,469 patients) (13,25). The median LOS in their sample was 5 days (IQR 2–10), which was longer compared to our cohort.

The main strengths of our study lie in the relatively large population of patients with disorders from the entire

spectrum of internal medicine, and the follow-up telephone interview regarding vital status and outcomes, which were assessed on day 30 after ED admission. Moreover, we had predefined inclusion of consecutive patients; therefore, patients were heterogeneous with regard to presenting chief complaint and main diagnosis, both of which allowed us to study different subgroups and assess robustness of our results.

Limitations

The present study nevertheless has several limitations. Because research in triage has been hampered by the lack of a universal gold standard for the real degree of urgency, the investigation of the validity of triage tools is limited to predictive validity and to surrogate markers, which are used instead for patient acuity. The closer and consistent integration of health information technology into triage systems might eventually assist in the development of a gold standard.

Although in this study we have reasonably endeavored to ensure a standardized assessment by different means,

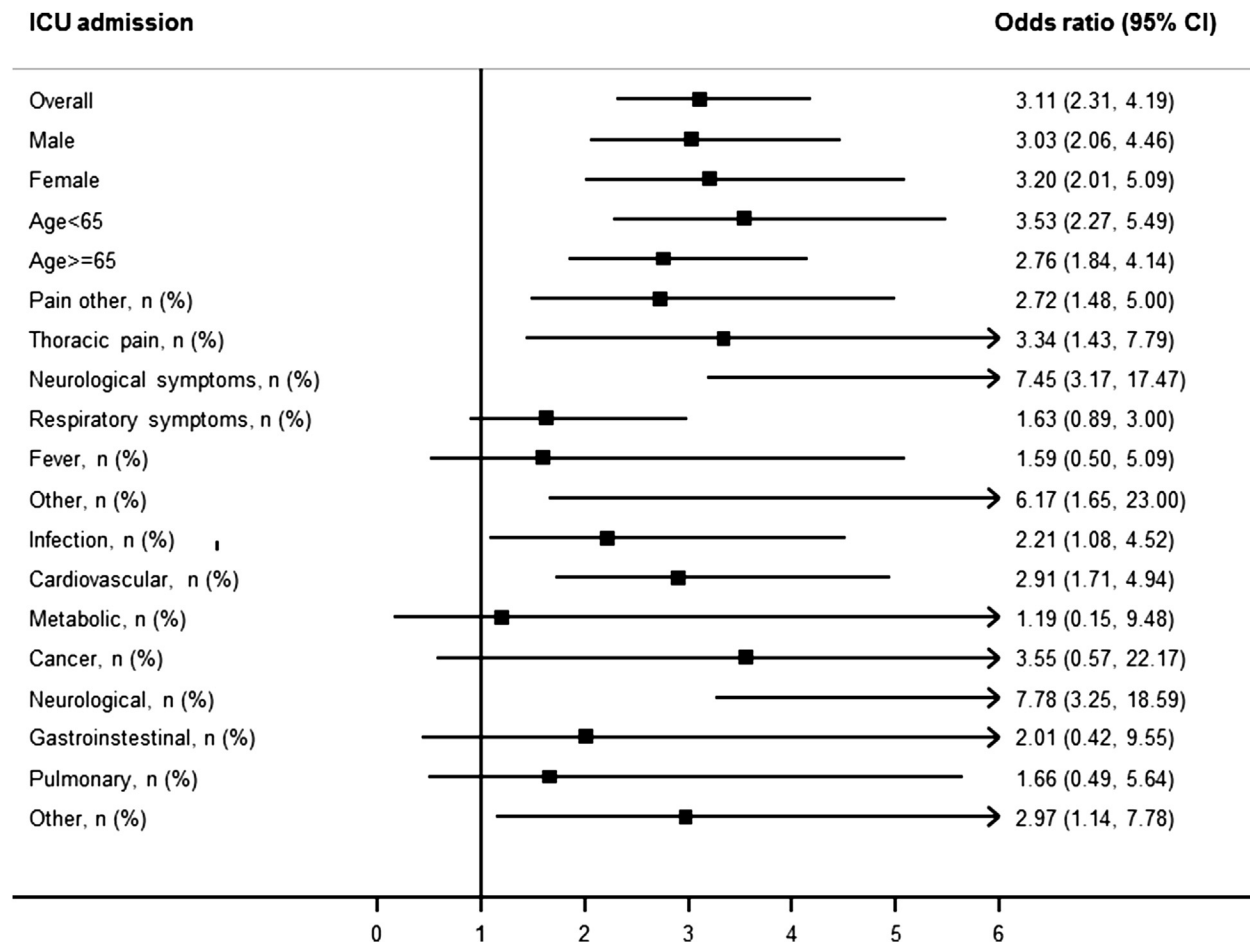


Figure 4. Forrest plot intensive care unit (ICU) admission. CI = confidence interval.

the lack of a universal gold standard remains a major limitation. We made every effort to minimize susceptibility to error and to give the adjudication as much objectivity as possible by blinding and by repeating the entire process at least two times, or in case of disagreement, three times. Nevertheless, the validity of the physician's assessment is limited by the methodology used to evaluate the assessment, and even more by the subjective nature of assessment itself: Even with a competent triage staff and the sophistication of diagnostic tools, judgments of urgency depend upon the triage staff's *perception* of urgency, which may differ from the *real* urgency of a patient's condition. Moreover, the patient's condition in the ED is not constant; symptoms can change even within a short period of time. Yet triage is based on an *initial* assessment of the degree of urgency, whereas the real degree of urgency could shift, even while the patient is still waiting in the ED. Finally, our study was limited to medical patients and performed at a single center; generalizability of the results is therefore limited due to the lack of an external validation cohort.

CONCLUSIONS

In a cohort of consecutive adult medical patients, the German version of the MTS in its second edition showed fair prognostic accuracy to predict high treatment priority, which is the assigned main task of a valid triage system. In addition, it showed fair performance in predicting ICU admission, whereas performance in predicting 30-day mortality was relatively poor. However, the addition of disease-specific prognostic biomarkers may increase its accuracy in the future.

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Schuetz, Mueller, Batschwaroff, Haubitz, and Faessler had the idea for the TRIAGE study and designed the study protocol. Steiner, Mueller, and Schuetz had the idea for this sub-study. Steiner, Renetseder, Laukemann, Kutz, Rast, and Felder helped in the data-gathering process. Buergi and Tobias provided information from the emergency department. Haubitz and Faessler were responsible for processing statistical data.

Steiner and Schuetz had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis, and they drafted the manuscript. All authors amended and commented on the manuscript revising it critically for important intellectual content. Byron

Anderson edited the manuscript. All authors read and approved the final manuscript. We are grateful to all local physicians, nursing staff, patients and patients' relatives participating in this study. We thank the medical clinic staff of the Cantonal Hospital Aarau for their very helpful assistance, patience, and technical support. We also extend our gratitude to all the physicians who contributed to data preparation, the Emergency Department staff, the study nurses of the TRIAGE study.

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ARTICLE SUMMARY

1. Why is this topic important?

The Manchester Triage System (MTS) is the most widely used scoring system in Europe, the UK, Australia, and other countries. Evaluation of its effectiveness is therefore instructive, for such evaluation can highlight features that possibly need improvement, not only in MTS, but also in triage systems generally.

2. What does this study attempt to show?

The study evaluated the effectiveness of MTS in accurately assessing degree of urgency and in predicting clinical outcomes in an unselected population of medical patients seeking treatment in an Emergency Department (ED).

3. What are the key findings?

The study found that MTS performed best in its assessment of treatment priority and prediction of mortality for patients presenting with neurological symptoms and disorders. However, its performance was fair only with respect to patients presenting with other symptoms, including infections, and cardiovascular and gastrointestinal complaints.

4. How is patient care impacted?

The inconsistent performance of MTS suggests that MTS's scoring system, and perhaps other triage scoring systems, may not provide an accurate assessment of treatment priority nor a reliable prediction of clinical outcomes for all patients who are admitted to the ED.